DIPARTIMENTO DI INGEGNERIA DELL'INFORMAZIONE

Explainable AI

Pietro Ducange, Francesco Marcelloni, Fabrizio Ruffini
AI Group
Department of Information Engineering
Largo Lucio Lazzarino 1
56123 PISA

E-mail: francesco.marcelloni@unipi.it

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Content of the lecture

- XAI models
- Ok, my AI-model works. But do I understand why and how?
  - Real-life scenarios
  - Stakeholders' motivations
- Trustworthiness
  - Transparency
    - Explainability
      - Interpretable by design models
      - Post-hoc techniques to explain opaque models' predictions

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eXplainable Artificial Intelligence (XAI) models

The research field of explainable artificial intelligence (XAI) addresses the explainability problems and methodologies (probably first introd. by McCarthy in '58 [1]).
Explainability (at that time): mostly related to requirements for a system "able to learn"

Now: different definitions, similar nuances of meaning. The underlying idea is to **understand how a system works**

The acronym **XAI** was then first used to indicate explainable artificial intelligence in 2004 by van Lent et. Al [2].

[1] J. McCarthy, "Programs with common sense,"
[2] M. van Lent, W. Fisher, and M. Mancuso, "An explainable artificial intelligence system for small-unit tactical behavior"

Why should I understand how my system works (in practice)?

1. Healthcare:
- **Diagnostic Systems:** Explainable AI helps doctors understand why a particular diagnosis or prediction was made, improving trust and aiding in decision-making.
- **Drug Discovery:** In pharmaceuticals, understanding how AI models arrive at conclusions about potential drug compounds is vital for validation and optimizing research efforts.

2. Finance:
- **Credit Scoring:** Explainability ensures fairness and transparency in credit decisions, allowing individuals to understand why they were granted or denied credit.
- **Risk Assessment:** Banks and financial institutions use AI for risk assessment; explainability helps in comprehending the factors influencing these assessments.

Why should I understand how my system works (in practice)?

3. Autonomous Vehicles:

- **Safety and Decision-Making:** Understanding the reasoning behind an autonomous vehicle's decision-making process is crucial for ensuring safety and building public trust in these technologies.

4. Criminal Justice:

- **Recidivism Prediction:** Explainability is crucial in predictive policing and recidivism prediction models to ensure fairness and prevent biases in decisions related to law enforcement and sentencing.

5. Ethical AI Deployment:

- **Bias Mitigation:** Transparent models enable the identification and mitigation of biases, promoting fairness and equity in various AI applications, such as hiring processes and resource allocation.

https://medium.com/@anshulgoel991/model-exploitability-using-shap-shapley-additive-explanations-and-lime-local-interpretable-cb4f5594fc1a

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Why should I understand how my system works (in practice)?

Goodness of results:
- Performance
- Bug fixing
- Augment results via expert-user domain-knowledge

UNDERSTANDING AI

Open the path to **new opportunities**:
- private and public services
- research fields

Accountability

Increase trust

Win reluctancy

Laws are respected

Easy end user feedback

Ethics and Fundamental rights are respected
- Bias fixing

Stakeholders:
(Domain) expert user
Data scientist
End-user
Regulators
High-tech Companies
Lawyers
Insurance companies

Increase interactivity -> increase interaction -> increase trust

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Why should I understand how my system works (in practice)?

- ....
- **Thinking "out of the human":**
  - it has been shown how AI-based systems are able not only to mimic the way humans come to a result, but also to create methods unlikely for the humans to follow.
    - AI is built by looking at human intelligence, but it is NOT identical
      - Wittgenstein's lion: ``If an intelligent lion could talk, we would not understand him'', because it could perceive the world in a qualitative different way with respect to humans *

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Trustworthy AI

Life-impacting decisions based on AI applications require trust

-> Regulators to set rules to ensure both wellbeing, fundamental rights preservation and societal development.

EU: commercial-and-fundamental rights oriented, trying to be at the forefront of legislations worldwide

USA: commercial-and-rights oriented. Different laws for different states

BRASIL

China: commercial oriented; focus on data privacy

Image: vector portal

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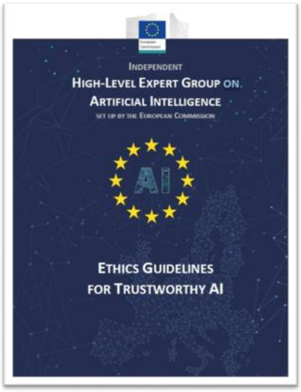
Trustworthy AI in EU

Life-impacting decisions based on AI applications require trust
Regulators to set rules to ensure both wellbeing and fund. rights.



In April 2019, the high-level expert group on AI set up by the EC presented the "**Ethic guidelines for trustworthy AI"**, according to which AI should be

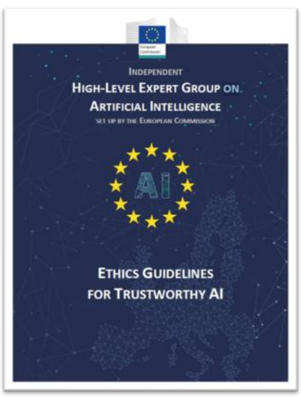
- **Lawful:** respecting all applicable laws and regulations
- **Ethical:** respecting ethical principles and values
- **Robust:** both from a technical and social perspective

AI act (EU first law on AI, based on the 2019 guidelines)

- April 2021 first proposal
- May 11, 2023: press release on draft negotiating mandate
- June 15, 2023: parliament vote for endorsement
- often referred to as the first law on AI

Does it matter? Next generation AI will have to comply with this type of legislation -> a gradual transition and advance planning is advisable.

Trustworthy AI in EU: ethical aspects



Seven key ethical requirements
evidenced by the expert group
Today: focus on TRANSPARENCY

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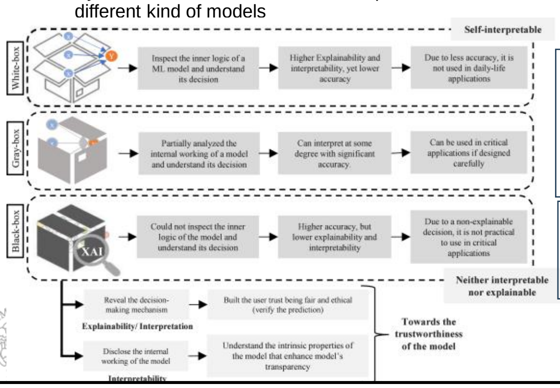
Transparency: 3 aspects

- Communication
  - Humans have the right to be informed that they are interacting with an AI system.
- Traceability
  - Data gathering/labelling and algorithms should be documented to the best possible standard
- **Explainability**
  - Systems and decisions should be explained in a manner adapted to the stakeholder concerned. Different explainability levels for different kind of models



Interpretable by design models

Linear models, Case-based reasoning models (KNN), Decision Trees

Fuzzy rule based models, Bayesian Networks

(traditional) ANN-based models (also DNN), ensemble models (eg: Random Forest)

Models needing post-hoc techniques to be explained

Image from Ali et al, "Explainable Artificial Intelligence (XAI): What we know and what is left to attain Trustworthy Artificial Intelligence"

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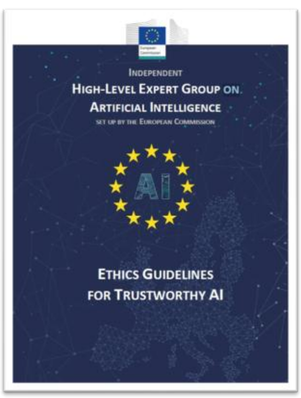
Trustworthy AI in EU: ethical aspects



…

Regarding Explainability:

Did you assess:

1. to what extent the decisions and hence the outcome made by the AI system can be understood?
2. to what degree the system's decision influences the organisation's decision-making processes?
3. why was this particular system deployed in this specific area?
4. what the system's business
   model is (for example, how does it create value for the organisation)?

(from https://www.europarl.europa.eu/cmsdata/196377/AI%20HLEG_Ethics%20Guidelines%20for%20Trustworthy%20AI.pdf ")

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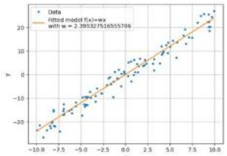
White (glass) boxes

Those models that are easily/ interpretable for a human.
Interpretability as: the ability to explain how decisions have been taken or to provide the meaning in a way that is understandable by a human. HOW a prediction has been taken

Linear/logistic regression models
Intuitively understandable: the larger the coefficient, the larger the impact of that feature on the output

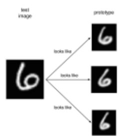
- Models are not complex -> low performance if the underlying phenomenon is complex



$f(x) = a_0 + a_1x_1 \, (+ a_2x_2 + \dots a_nx_n)$

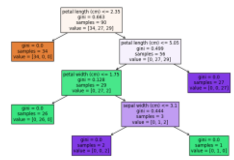
Case based reasoning
Humans intuitively look in the past for similar input situations, and use past outputs for new inference

- typical methods involve k-NN (comput. expensive) or defining Prototypes (which is a good prototype?)



Decision tree models
Similar to human reasoning as a sequence of simple decisions in the form “if-then” (logical models)

- Simple univariate thresholds
- Usually good performance
- Prone to overfitting



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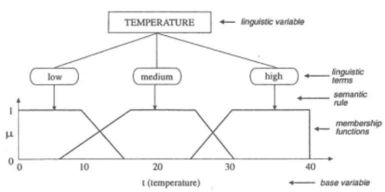
Grey boxes

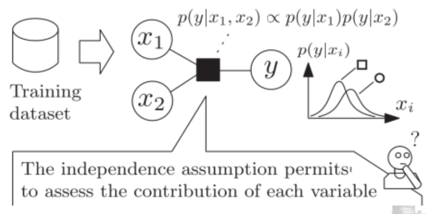
Those models that users can interpret at some degree if they are carefully designed [1]

Fuzzy-rule based systems: If-then-rules
Linguistic rules and fuzziness help interpretability



If temperature is "low" **and** humidity is "medium"
Then tourist_number = 0.25 +0.4*temperature-0.2*humidity (or tourist number is "high")

Bayesian Models
usually takes the form of a probabilistic directed acyclic graphical model whose links represent the conditional dependencies between a set of variables. For example, a Bayesian network could represent the probabilistic relationships between diseases and symptoms



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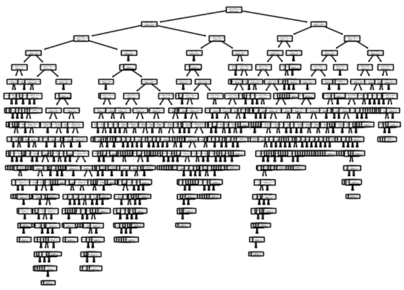
Focus on Decision Trees and FRBS

Both in the category of "logical models"
Both can be expressed by a set of “if-then" rules

Figs from A. B. Arrieta et al,, "Explainable Artificial Intelligence (AI): Concepts, taxonomies, opportunities and challenges toward responsible AI", 2020
[1] Ali et al, "Explainable Artificial Intelligence (XAI): What we know and what is left to attain Trustworthy Artificial Intelligence"

How to evaluate interpretability?

1. **Quantitative evaluation (complexity of a system as a proxy for interpretability):**
   - Number of rules / number of antecedents / number of coefficients (for TSK-FRBS)



DecisionTree for diabetes regression task (see nb)
Features = 10

```
from sklearn.tree import export_text
tree_rules = export_text(DT_reg, feature_names=list(X_train.columns))
print(tree_rules)
```

```
|--- s5 <= 0.02
|   |--- bmi <= 0.01
|   |   |--- s3 <= 0.02
|   |   |   |--- s5 <= -0.04
|   |   |   |   |--- s6 <= -0.00
|   |   |   |   |   |--- s6 <= -0.08
|   |   |   |   |   |   |--- bmi <= -0.06
|   |   |   |   |   |   |   |--- s3 <= 0.01
|   |   |   |   |   |   |   |   |--- value: [134.00]
|   |   |   |   |   |   |   |--- s3 >  0.01
|   |   |   |   |   |   |   |   |--- value: [158.00]
|   |   |   |   |   |   |--- bmi >  -0.06
|   |   |   |   |   |   |   |--- value: [70.00]
|   |   |   |   |   |--- s6 >  -0.08
|   |   |   |   |   |   |--- bmi <= -0.08
|   |   |   |   |   |   |   |--- value: [116.00]
|   |   |   |   |   |   |--- bmi >  -0.08
|   |   |   |   |   |   |   |--- s2 <= 0.02
|   |   |   |   |   |   |   |   |--- s1 <= -0.08
|   |   |   |   |   |   |   |   |   |--- bmi <= -0.05
|   |   |   |   |   |   |   |   |   |   |--- s6 <= -0.05
|   |   |   |   |   |   |   |   |   |   |   |--- value: [88.00]
|   |   |   |   |   |   |   |   |   |   |--- s6 >  -0.05
|   |   |   |   |   |   |   |   |   |   |   |--- value: [83.00]
|   |   |   |   |   |   |   |   |   |--- bmi >  -0.05
|   |   |   |   |   |   |   |   |   |   |--- value: [71.00]
|   |   |   |   |   |   |   |   |--- s1 >  -0.08
|   |   |   |   |   |   |   |   |   |--- s6 <= -0.06
|   |   |   |   |   |   |   |   |   |   |--- s2 <= -0.04
|   |   |   |   |   |   |   |   |   |   |   |--- value: [65.00]
|   |   |   |   |   |   |   |   |   |   |--- s2 >  -0.04
|   |   |   |   |   |   |   |   |   |   |   |--- value: [78.00]
|   |   |   |   |   |   |   |   |   |--- s6 >  -0.06
|   |   |   |   |   |   |   |   |   |   |--- bmi <= -0.02
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 6
|   |   |   |   |   |   |   |   |   |   |--- bmi >  -0.02
|   |   |   |   |   |   |   |   |   |   |   |--- value: [31.00]
|   |   |   |   |   |   |   |--- s2 >  0.02
|   |   |   |   |   |   |   |   |--- value: [113.00]
|   |   |   |   |--- s6 >  -0.00
|   |   |   |   |   |--- s2 <= -0.00
|   |   |   |   |   |   |--- s2 <= -0.02
```

…. and many others

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Focus on Decision Trees and FRBS

Both in the category of "logical models"
Both can be expressed by a set of “if-then" rules

How to evaluate interpretability?

1. **Quantitative evaluation:**
   - Number of rules / number of antecedents / number of coefficients (for TSK-FRBS)
2. **Qualitative evaluation:**
   - Semantic
     - for me, low temperature is from 0->5 Celsius degrees; but for you?
     - If I use many fuzzy sets, I can end up with many linguistic labels such as "very very very hot" != "very very hot"… (<- is it really an interpretable system???)

NOTE on Sparsity: humans can(?) handle 7±2 cognitive entities at the same time. [1,2]
-> Less is okay(?): sparse scenarios could be well handled by “not complex” logical models, allowing maximum interpretability

[1] G. A. Miller "The magical number seven, plus or minus two: Some limits on our capacity for processing information.", 1956
[2] T.L.Saaty "Why the magic number seven plus or minus two", 2003

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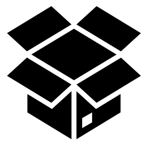
Black (opaque) boxes

Those models that are not especially designed to be interpretable by humans (still, we want them to be explainable)

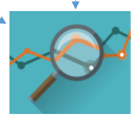
Explainability as: **given a certain audience, explainability refers to the details and reasons a model gives to make its functioning clear or easy to understand.** [A.B. Arrieta et al, "Explainable artificial intelligence (xai): Concepts, taxonomies, opportunities and challenges toward responsible ai"]

Why does a model predict?

Dataset X



Opaque model m



Explanator model e

Prediction of the model m
\hat(y)= m(x)

Explanation given an input and a model
expl = e(m,x)

USER

NOTE: **nomenclature** is not universally accepted.
Actually, we are in the era of "naming" things
Image: vector portal

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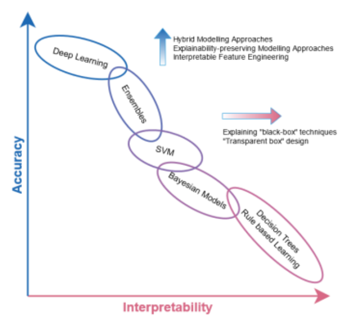
XAI: Explainable AI

How to **achieve** explainability?

- Design of **Inherently Interpretable Models** (here: white and grey boxes (see previous slides))
- **Post-hoc explainability** techniques for black-boxes (opaque models)

Taxonomy:
4(?) points of view



Post-hoc methods: Resemblance with human approach	Post-hoc methods: By dataset	Post-hoc methods: Global / local	Post-hoc methods: Model agnostic / model specific

Image from Bodria et al., "Benchmarking and survey of explanation methods for black box models"

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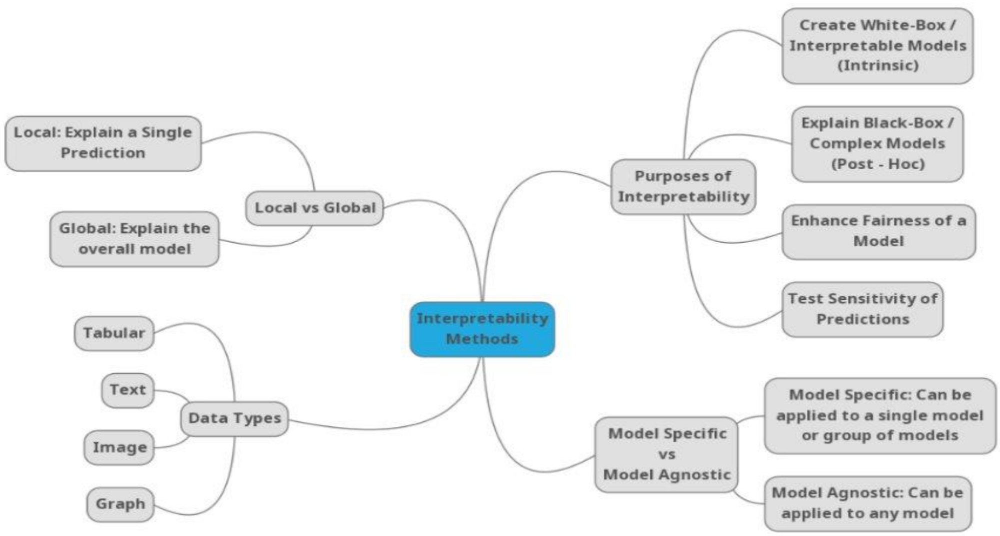
Another point of view



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Post-hoc methods resembling human approach

Schematically:

- Text explanations
- Visual explanations
- Local explanations
- Explanations by example
- Explanations by simplification -> surrogate models.
- Feature relevance explanations
- … (more incoming)

[A.B. Arrieta et al, "Explainable artificial intelligence (xai): Concepts, taxonomies, opportunities and challenges toward responsible ai"]

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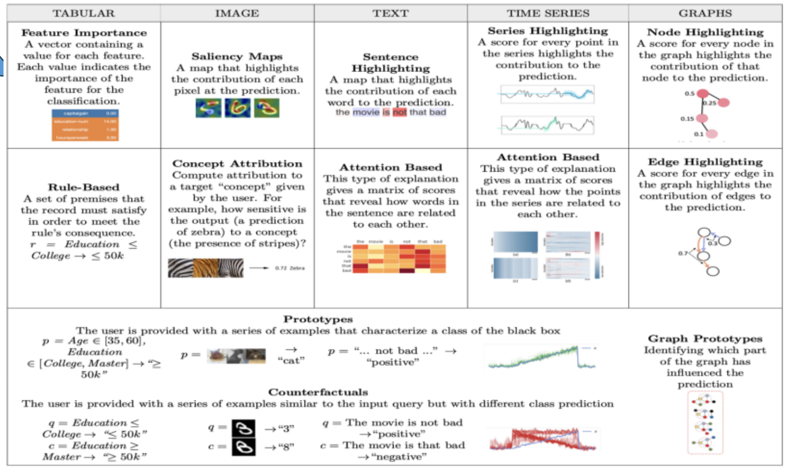
Post-hoc methods by dataset

How to **achieve** explainability?

- Design of **Inherently Interpretable Models**
  (see previous slide = white and grey boxes)
- **Post-hoc explainability** techniques for black-boxes (opaque models)

By dataset



TABULAR	IMAGE	TEXT	TIME SERIES	GRAPHS
Feature Importance A vector containing a value for each feature. Each value indicates the importance of the feature for the classification.	**Saliency Maps** A map that highlights the contribution of each pixel at the prediction.	**Sentence Highlighting** A map that highlights the contribution of each word to the prediction. the movie is not that bad	**Series Highlighting** A score for every point in the series highlights the contribution to the prediction.	**Node Highlighting** A score for every node in the graph highlights the contribution of that node to the prediction.
Rule-Based A set of premises that the record must satisfy in order to meet the rule's consequence. $r = Education \leq College \rightarrow \leq 50k$	**Concept Attribution** Compute attribution to a target "concept" given by the user. For example, how sensitive is the output (a prediction of zebra) to a concept (the presence of stripes)?	**Attention Based** This type of explanation gives a matrix of scores that reveal how words in the sentence are related to each other.	**Attention Based** This type of explanation gives a matrix of scores that reveal how the points in the series are related to each other.	**Edge Highlighting** A score for every edge in the graph highlights the contribution of edges to the prediction.
Prototypes The user is provided with a series of examples that characterize a class of the black box: $p = Age \in [35, 60]$, $Education \in [College, Master] \rightarrow$ "$\geq 50k$"; $p =$ [images] $\rightarrow$ "cat"; $p =$ "... not bad ..." $\rightarrow$ "positive"				**Graph Prototypes** Identifying which part of the graph has influenced the prediction
Counterfactuals The user is provided with a series of examples similar to the input query but with different class prediction: $q = Education \leq College \rightarrow$ "$\leq 50k$", $c = Education \geq Master \rightarrow$ "$\geq 50k$"; $q =$ [image] $\rightarrow$ "3", $c =$ [image] $\rightarrow$ "8"; $q =$ The movie is not bad $\rightarrow$ "positive", $c =$ The movie is that bad $\rightarrow$ "negative"				

Image from Bodria et al., "Benchmarking and survey of explanation methods for black box models"

Model agnostic/ model specific post-hoc methods

Model specific post-hoc method:

- Meant to explain a specific AI model

PROS:
-> tailored for a specific model

CONS:
Cannot be applied to all models

Model agnostic:

- not tied to any specific AI model

PROS:
-> high flexibility

CONS:
-> possibly high computational effort

Global and Local explainability and interpretability

Post-hoc explainability:

- **Local explanators:** explain individual predictions of the AI model
- **Global explanators:** provide global insights into the entire model.
  - Often comes from aggregation of local explainations

Inherently interpretable:

- **Local interpretability:** refers how the model is defined for a specific region (eg: for a Rule Based System, the activated regions and their rules)
- **Global interpretability**: (schematically): it refers to the structure of the model -> complexity notion (eg: number of rules for extracted from a decision tree)

Post-hoc techniques intuition

Given a dataset with j=1,…, J features and a learned model model *f*
We compute a score metric (eg. accuracy, MAE...) ***s*** of f

To evaluate the importance of a single feature *j*, it is intuitive to see how the metric score changes while the feature value changes. If the score changes a lot, we can intuitively say that this feature is important

Thus,

1. for each feature j we randomly shuffle K times (k=1,2,…, K) its value (-> we obtain a "corrupted version of the data")
2. we compute the score on corrupted data s(k,j)
3. Compute the importance of the specific feature I(j) as $i(j) = s - \frac{1}{K}\sum_{k=1}^{K} s(k,j)$

- **Question: if your feature is constant -> the values are all the same, what is the importance I(j) of this feature? -> it would be ???**

Extracted from W. Pedrycz course material

Post-hoc techniques intuition

Given a dataset with j=1,..., J features and a learned model model *f*
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- **Question: if your feature is constant -> the values are all the same, what is the importance I(j) of this feature? -> it would be = 0 … no importance**

Extracted from W. Pedrycz course material

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SHAP: intuition and background

SHAP[1] is a post-hoc method for estimating the Shapley[2] values.
Intuition: soccer team as a realization of cooperative game theory

Imagine a soccer team wins a match 3-0.
How much did each player contribute to the victory?

Not only the goal scorer matters: defenders, midfielders, and pass creators all play a role.

The team = the model | Players = features | Victory = prediction

Shapley values consider:
All possible orders (coalitions) in which players (features) are added
The change in team performance (prediction) with each addition
The average contribution of each player across all combinations
Fairly attributes the prediction to each feature, just like crediting players in a team win.

1: Lundberg et al. A unified approach to interpreting model predictions (2017)
2: Shapley values introduced by Lloyd Shapley, Nobel Memorial Prize in Economics in 2012]

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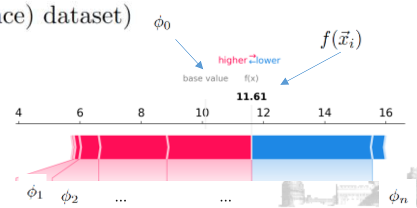
SHAP: from game theory to XAI

Replace **players** with **features**
Replace **profit** with model **prediction**
Shapley values quantify the impact of each feature on model prediction

$$\hat{y}_i = f(\vec{x}_i) = \phi_0 + \sum_{j=1}^{F} \phi_j$$

- f : Predictive model
- $\vec{x}_i$: i^{th} input instance, with dimension F
- ϕ_0 : base value (model f predictions averaged over a background(reference) dataset)
- ϕ_i : Shapley values

To evaluate each coalition, we:
- Keep only the selected features from the instance x
- Replace the other features using values from a background (reference) dataset
- This simulates what the model would predict without access to certain features.



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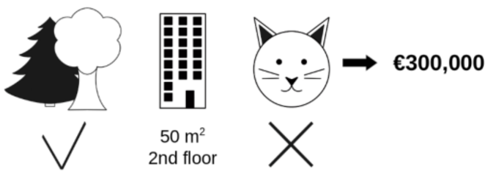
Shapley: math

You have trained an AI-model to predict apartment prices.
For an individual instance it predicts €300,000 and you need to explain this prediction.
The apartment has an area of 50 m^2, is located on the 2nd floor, has a park nearby and cats are banned. The average prediction for all apartments is €310,000



How much has each feature value contributed to the prediction compared to the average prediction?

Molnar, "Interpretable Machine Learning"

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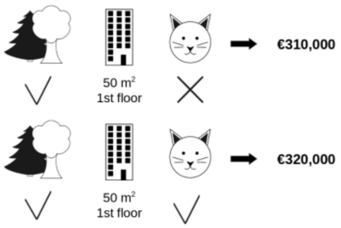
Shapley: math

The Shapley value is the average marginal contribution of a feature value across all possible coalitions (here: coalition = groups of players=features).
Example: we want to know the Shapley value for the feature "cat_banned"



Coalition 1: same as instance to be predicted, except for 1st floor -> 2nd floor. Model prediction = 310,000

Coalition 2: same as instance to be predicted, except for 1st floor -> 2nd floor and cat is allowed.
Model prediction = 320,000

Marginal contrib of cat_allowed = **10,000** = 320,000-310,000

We repeat this computation for all possible coalitions -> we calculate the average

Molnar, "Interpretable Machine Learning"

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SHAP package: tabular example with breast cancer (0 = diseased, 1=healthy)

```
import pandas as pd
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
import shap

seed = 42
# Load the Breast Cancer dataset
cancer = load_breast_cancer()
X = pd.DataFrame(cancer.data, columns=cancer.feature_names)
y = pd.DataFrame(cancer.target, columns=['target'])

# Split the dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=seed)

# Initialize and train RandomForestClassifier
rf_classifier = RandomForestClassifier(random_state=seed)
rf_classifier.fit(X_train, y_train.values.ravel())

# Explain the model's predictions using SHAP
explainer = shap.TreeExplainer(rf_classifier)

# Calculate SHAP values for the entire test set
shap_values = explainer.shap_values(X_test)
```

>> pip install shap
or
>> conda install -c conda-forge shap
https://shap.readthedocs.io/en/latest/index.html

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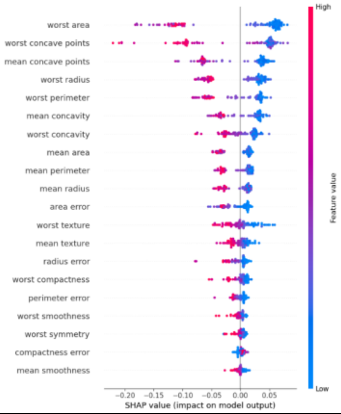
SHAP package: tabular example with breast cancer (0 = diseased, 1=healthy)

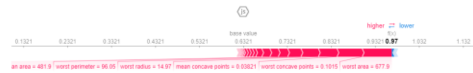
Beeswarm summary plot
-> many single predictions explanations -> global overview (global explanation)

```
shap.summary_plot(shap_values[1], X_test, show=False)
plt.show()
```



Force plot: single prediction explanation (-> local explanation)

```
# Visualize a single prediction
shap.initjs()
shap.force_plot(explainer.expected_value[1], shap_values[1][0, :], X_test.iloc[0, :])
```



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Francesco Marcelloni

LIME: Local Interpretable model-agnostic exploration

We provide a **simpler** model trying to mimic the model **locally**

Steps:
1. We consider the model *m* (usually opaque) we want to explain and an input instance *I*
2. We create perturbed samples "close" to the instance I
   - On this sample (that is "synthetic"), we use the model "m" to get the predictions
3. We train a simpler model, interpretable by design, on the couple
   - Synthetic inputs -> model "m" predictions (there is a weight based on the distance) from "i"
4. The simpler model (also called surrogate model), being interpretable by design (such as a linear model or a decision tree) is used to explain the model "m" locally

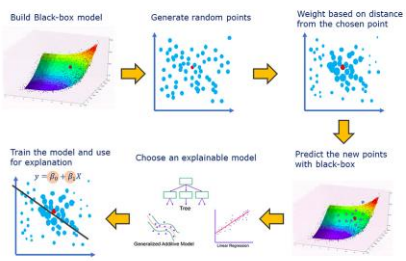

image from Giorgio Visani

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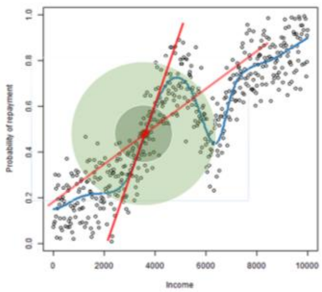
LIME: Local Interpretable model-agnostic exploration

PRO:
Easy to understand
CONS:
- Synth data generation: if not careful, can create "not accurate" local models?
- Notion of distance: how to chose? How to stop generating synth data?
- It is a local model -> you could "patch" many local models -> is it feasible
- When evaluating the performance of the full strategy, you must take into account the double errors in the prediction -> the errors related to model "m" and the errors related to the surrogate model.



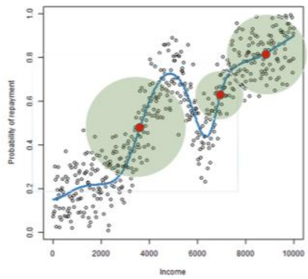


image from Giorgio Visani

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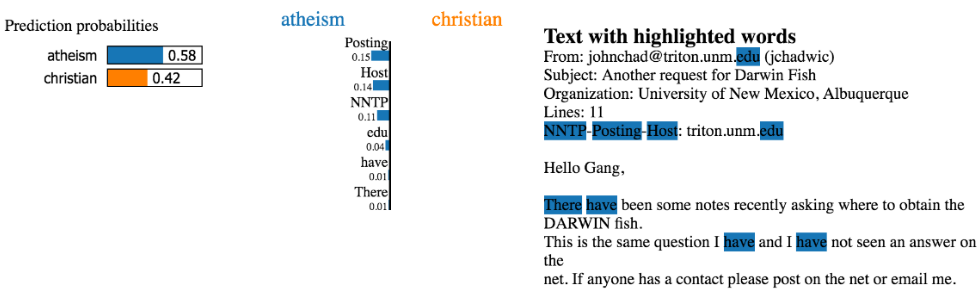
LIME: package

\>> pip install lime
https://github.com/marcotcr/lime

Two class case, text

Negative (blue) words indicate atheism, while positive (orange) words indicate christian. The way to interpret the weights by applying them to the prediction probabilities. For example, if we remove the words Host and NNTP from the document, we expect the classifier to predict atheism with probability 0.58 - 0.14 - 0.11 = 0.31.



m Giorgio Visani

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Counterfactual explanation

Example:
You are taking an exam, but you do not pass.

Intuitive consequent questions:
-> Why?
-> What change should have been done so that my request has been accepted?

Student x0: I studied 20 hours + read 2 books + answered last year 10 questions + 99 aperitifs

Given the output y0 and the corresponding input x0, what should be different in x0 to get another output y1?

-> in general, I'm very interested in the SMALLEST change in x0

-> Student x0 + some_small_changes: I studied X hours + read Y books + answered last year Z questions + gone to T aperitifs

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Counterfactual explanation

Example:
You are taking an exam, but you do not pass.

Intuitive consequent questions:
-> Why?
-> What change should have been done so that my request has been accepted?

Student x0: I studied 20 hours + read 2 books + answered last year 10 questions + 99 aperitifs

Given the output y0 and the corresponding input x0, what should be different in x0 to get another output y1?

-> in general, I'm very interested in the SMALLEST change in x0

-> Student x0 + some_small_changes: I studied 40000 hours + read 10 books + answered last year 20 questions + gone to 2 aperitifs

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Counterfactual explanation: packages

Python packages:

Alibi: https://docs.seldon.io/projects/alibi/en/stable/index.html

Mace: https://github.com/charmlab/mace

Dice: https://github.com/interpretml/DiCE

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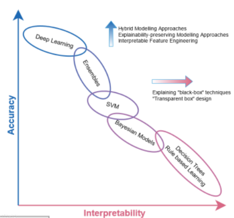
Focus on Images

How to **achieve** explainability?

- Design of **Inherently Interpretable Models**
- **Post-hoc explainability** techniques for black-boxes (opaque models)



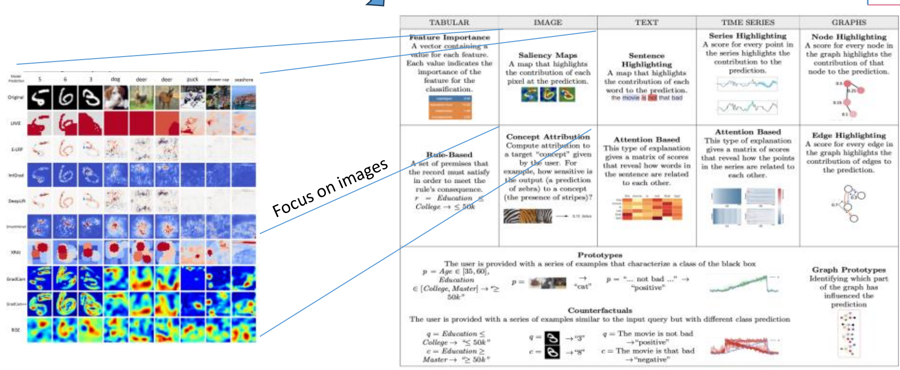


Image from Bodria et al., "Benchmarking and survey of explanation methods for black box models"

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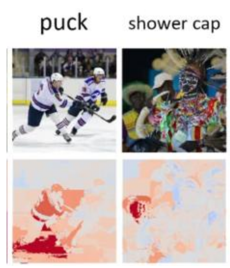
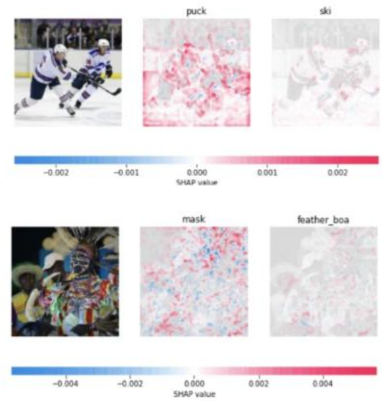
SHAP and LIME for images

A *Saliency Map (SM)* is an image in which a **pixel's brightness** represents how **salient** (important) the pixel is for the **model decision**.
In a classification task, a positive value means that that pixel has contributed positively to the classification, while a negative one means that it has contributed negatively.
(intuition: for example, in the plots on the right: red = positive, blue=negative)

-> Saliency maps for images is based on the same intuition of "importance feature" for tabular and text data



LIME is a local post-hoc model-agnostic explainer, using superpixels (segments of the image)



SHAP, in the version GRAD-shap, produces a Saliency map for each class in the input image can explain simultaneously different labels

Images from Bodria et al., "Benchmarking and survey of explanation methods for black box models"

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A final note

In literature (and also in the public debate) you will find that the **nomenclature is not yet fixed.**
So, terms such as interpretability, explainability, transparency are sometimes used as synonyms.

We have to wait for the community to reach "**nomenclature maturity**"

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